Achyut Niroula

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EXPERIENCE

Norlease | northlease.ca — Aug. 2026 – present
Founder — North Bay, ON

- Applied a direct cost-benefit lesson from prior AWS GPU infrastructure experience to choose Hetzner's flat-rate hosting over metered cloud billing, reducing infrastructure cost by an estimated 80-90 percent for a system with no GPU requirement.
- Personally met with approximately 20 prospective landlords in person, at their homes, to explain the platform and secure early onboarding ahead of public launch.
- Built a full-stack application end to end using React and TypeScript on the frontend and a Python/FastAPI backend, designing every architectural decision from data model through deployment independently.
- Designed and built RESTful APIs integrating multiple third-party services, working directly with SQL and a relational PostgreSQL schema throughout.
- Set up CI-style deployment and containerized infrastructure through Docker and Coolify, applying automated deployment flows from the earliest stage of the project.
- Wrote extensive automated tests, including Playwright end-to-end suites run against real production infrastructure, embedding testing into every release.

Nipissing University — Jan. – Apr. 2026
Honours Research — North Bay, ON

- Built a multi-model AI video perception pipeline on AWS (EC2 GPU instances, S3, SQS, DynamoDB), running the compute-intensive worker roughly 30 hours per week and manually managing instance state to control cost on a metered, per-hour GPU billing model.
- Selected AWS specifically for hands-on experience with enterprise cloud infrastructure and GPU compute access, gaining direct insight into the cost trade-offs of metered cloud billing versus flat-rate hosting for compute-bound workloads.

Nipissing University — May–Aug. 2023, 2024, 2025
Research Assistant & Web Developer — North Bay, ON

- Built and maintained a full-stack platform across three consecutive summer contracts, collaborating directly with a stakeholder to extract requirements and translate them into working features.

PROJECTS

LEON | github.com/achyutniroula | *Java, Spring Boot, Python, FastAPI, React, PostgreSQL, pgvector*

- Built the backend in Java using Spring Boot, then migrated to Python/FastAPI for async processing, working with React on the frontend throughout and applying algorithmic thinking to a RAG-based retrieval system.

JobHunt AI + ResOptimum | github.com/achyutniroula | *Python, FastAPI, React, TypeScript, Docker*

- Built a distributed application with a React/TypeScript frontend and RESTful FastAPI backend, applying full pytest coverage across every endpoint and containerized deployment with Docker Compose.

LEADERSHIP

Nipissing University — 2022 – 2025
Computer Science Student Representative — North Bay, ON

- Served as the primary liaison between students and the Dean for three consecutive years, ensuring concerns were documented and formally addressed.
- Organized hackathons, seminars, and workshops from the ground up, managing sponsor and faculty relationships, logistics, and turnout for each event.

EDUCATION

Nipissing University — May 2026
B.Sc. Honours Specialization in Computer Science — GPA: 3.7/4.0 — North Bay, ON

- Awards: President's Scholarship (2021), Carl Sanders Scholarship (2022–2025)
- Relevant coursework: Software Engineering (89%), Databases & Data Management (96%), Distributed Systems (96%)

TECHNICAL SKILLS

Frontend: React, TypeScript, JavaScript, HTML/CSS
Backend: Python (directly transferable to Node.js), FastAPI, Flask, Spring Boot, RESTful APIs
Databases: PostgreSQL, SQL query design and architecture, pgvector, Redis, SQLite
Testing & DevOps: pytest, JUnit, Playwright, automated testing, Docker, CI/CD-aligned deployment
Other: Software architecture, algorithms, Git, technical specification extraction from requirements